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-- DROP DATABASE if it exists to start fresh
-- IF DB_ID('SCHOOL') IS NOT NULL
-- DROP DATABASE SCHOOL;
-- GO

-- CREATE DATABASE and switch to it
CREATE DATABASE SCHOOL;
GO
USE SCHOOL;
GO

-- CREATE TABLE Accounts
CREATE TABLE Accounts (
    AccountID INT PRIMARY KEY,      -- Unique ID for each account
    AccountName NVARCHAR(100),      -- Account holder's name
    Balance DECIMAL(10,2)           -- Money: up to 10 digits, 2 decimal places
);
GO

-- INSERT initial accounts
INSERT INTO Accounts (AccountID, AccountName, Balance) VALUES
(1, 'Alice', 1000.00),
(2, 'Bob', 500.00);
GO

-- Show starting balances
SELECT * FROM Accounts;
GO

-- SIMPLE TRANSACTION: Move 200 from Bob to Alice
BEGIN TRANSACTION; -- Start a "safe box" for changes
UPDATE Accounts
SET Balance = Balance - 200.00
WHERE AccountID = 2; -- we now proceed to subtract 200 from Bob
UPDATE Accounts
SET Balance = Balance + 200.00
WHERE AccountID = 1; -- the 200 deducted from Bob to be added to Alice
COMMIT TRANSACTION; -- if the transaction is successful save changes permanently
GO
SELECT * FROM Accounts;
GO

-- TRY/CATCH TRANSACTION: Let try adding big money. To perfectly execute this one,
you must select the entire TR/CATCH BLOCK
BEGIN TRANSACTION;
BEGIN TRY
    UPDATE Accounts
    SET Balance = Balance + 6930700.00
    WHERE AccountID = 2; -- Add big amount to Bob

    UPDATE Accounts
    SET Balance = Balance + 400.00
    WHERE AccountID = 1; -- Add 400 to Alice

    COMMIT TRANSACTION; -- Save if no errors
    SELECT * FROM Accounts;
END TRY
BEGIN CATCH
    ROLLBACK TRANSACTION; -- Undo all changes if error occurs
    DECLARE @ErrorMessage NVARCHAR(4000) = ERROR_MESSAGE(); -- Get error message
    RAISERROR(@ErrorMessage, 16, 1); -- Show the error
END CATCH;
GO

-- FORCE ERROR TRANSACTION: insert text into number column
BEGIN TRANSACTION;
BEGIN TRY
    UPDATE Accounts
    SET Balance = 'NotANumber' -- Will probably fail because we defined our balance
    integers not text
    WHERE AccountID = 2;

    COMMIT TRANSACTION; -- Would save if no error
    SELECT * FROM Accounts; -- (balances won't be affected because NotANumber is not
an integer)
END TRY
BEGIN CATCH
    ROLLBACK TRANSACTION; -- Undo changes
    PRINT 'Rollback occurred: ' + ERROR_MESSAGE(); -- Show error
END CATCH;
GO

-- if rollback occurred successfully then we expect balances from TRY/CATCH to
display as the final balance on Alice and Bob's accounts
SELECT * FROM Accounts; -- Confirm final balances not affected
GO

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